

VINAY KUMAR

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SUMMARY

Software Engineer specializing in backend development, distributed systems, and scalable full-stack applications. Experienced in building REST APIs, real-time systems, and cloud-native applications using Node.js, Express.js, Next.js, TypeScript, MongoDB, and PostgreSQL. Dedicated to solving complex engineering problems through end-to-end ownership, continuous learning, and collaborative problem-solving to deliver reliable, high-performance software.

EDUCATION

VIT BHOPAL UNIVERSITY

Bachelor of Technology, Computer Science & Engineering (AI & ML)

Cumulative GPA: 8.92

Bhopal, MP

Sept 2023 - Present

PROJECTS

VeggieMap — Hyperlocal Vegetable Vendor Finder

[GitHub](#) | [Live](#)

- Engineered a real-time hyperlocal marketplace allowing users to track vegetable vendors, designing a scalable WebSocket architecture leveraging 5 km geospatial room partitioning to efficiently broadcast location updates only to nearby users.
- Implemented a **5-second controlled latency** for location broadcasts using Socket.io to reduce unnecessary network traffic and backend processing while maintaining a responsive UI, and configured an Nginx reverse proxy on AWS EC2 to efficiently load-balance traffic and ensure high availability.

SMS Gateway — Scalable Android SMS Infrastructure

[GitHub](#) | [Live](#)

- Built an Android-based SMS gateway for reliable OTP and campaign delivery via cellular plans, implementing strict Rate Limiting (3 SMS/hour per user) to protect network integrity and eliminate recurring third-party SMS service costs.
- Developed the backend infrastructure using Node.js, BullMQ, and a Redis cache for reliable message queuing, integrating Firebase Cloud Messaging (FCM) to enable reliable real-time message synchronization between the backend and Android clients, achieving **<2.5 sec end-to-end latency**.

WebPull AI — AI-Powered Documentation Scraper

[GitHub](#) | [Live](#)

- Developed a full-stack documentation extraction platform using Next.js, Express.js, and Crawlee to scrape dynamic websites and convert content into structured Markdown for seamless integration into external knowledge bases.
- Integrated the Google Gemini API to automate URL discovery and generate AI-assisted documentation summaries when official documentation is unavailable, utilizing Server-Sent Events (SSE) and JSZip to package and stream results in real time.

TECHNICAL SKILLS

Languages:	C++, TypeScript, JavaScript, Python
Backend:	Node.js, Express.js, REST APIs, Socket.io (WebSockets), JWT Authentication, Mongoose, BullMQ, Redis
Frontend:	React, Next.js, Tailwind CSS, HTML, CSS
Databases:	MongoDB, PostgreSQL, Firebase
Cloud & DevOps:	AWS EC2, Docker, Linux, Nginx, PM2, Git, GitHub Actions, SSH
Machine Learning:	NumPy, Pandas, Scikit-learn, XGBoost, ONNX

CERTIFICATIONS

AWS Certified Cloud Practitioner — Amazon Web Services

[Link](#)

Cloud Computing — NPTEL, IIT Madras

ACHIEVEMENTS & ACTIVITIES

- Coding:** Solved 250+ algorithmic problems on LeetCode covering arrays, graphs, trees, dynamic programming, and binary search.
- Leadership:** Active member of the PR and Outreach team for the university's Health-O-Tech club, managing community engagement.
- Technical Migration:** Successfully migrated a production deployment from Oracle Cloud Infrastructure (OCI) to AWS EC2 by provisioning the new environment, validating application stability, and switching DNS records to ensure uninterrupted service.